Hierarchical Projection as a Validity Layer for Multilingual ESG Promise Verification

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12.55% → 0 invalid tuples, by construction

△ wF1 −0.001 official score, undetected (p_{Holm}=1.000)

tuple acc. +0.035 whole-tuple accuracy (p_{Holm}=.001)

32/32 vs 14/32 external arms improving: tuple acc. vs wF1

1 · Problem: plausible fields, invalid tuple

- Four fields describe one ESG claim: promise status (PS), verification timeline (VT), evidence status (ES), evidence quality (EQ).
- Hard implications: PS = No forces VT = ES = EQ = N/A, and ES = No forces EQ = N/A.
- Only **17 of 120** label combinations are legal.
- Independent argmax emits invalid tuples on **12.55%** of Chinese rows.
- The official metric scores each field on its own, so an invalid tuple still keeps **44.3%** of the weighted field credit.

2 · Method: seven decision rules

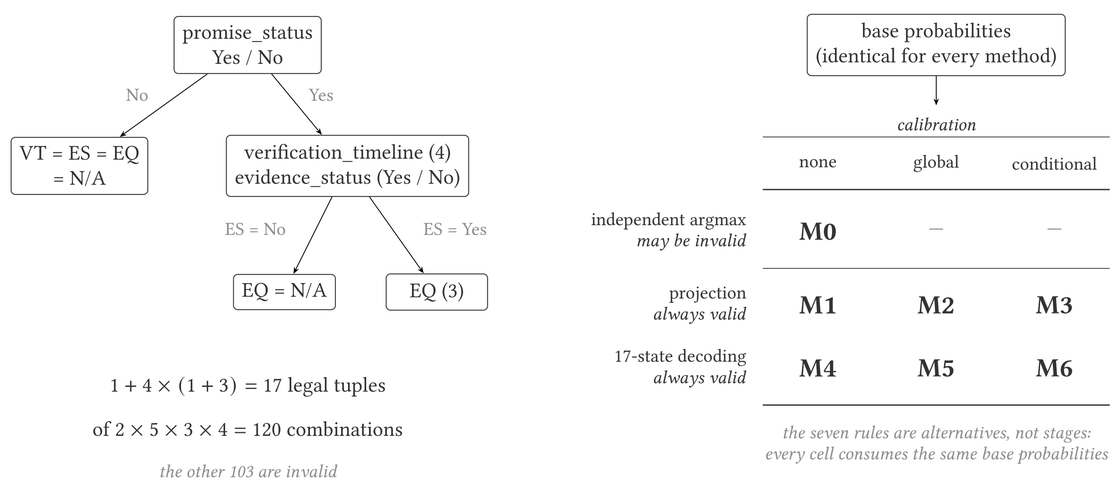


Figure 1 of the paper: the label hierarchy and the seven alternative decision rules

- One fine-tuned encoder with four independent heads; every rule reads the same cross-fitted probabilities.
- **Projection** (greedy, top-down): pick PS; under PS = Yes pick the best substantive VT and ES; under ES = Yes the best substantive EQ. Always legal, no retraining.
- **17-state decoding**: return the legal state with the highest summed log-score.
- Calibration: none, global, or conditional class biases. M0 is independent argmax.

3 · Setup

- Chinese anchor: 2,000 paragraphs from 49 reports; `hfl/chinese-roberta-wwm-ext-large`.
- Five-way cross-fitting, three seeds, two protocols (document-disjoint primary).
- Paired report-cluster bootstrap, Holm-adjusted within each pre-specified family.
- External: English, French, Japanese (400 paragraphs each), Korean (500 PDF pages); 4 backbones × $\lambda \in \{0, 0.3\}$ = **32 arms**.

4 · Results: Chinese anchor

ID	Calibration	Decoding	PS	VT	ES	EQ	wF1 (official)	Tuple acc.	Invalid %
M0	None	Independent	0.796	0.464	0.650	0.424	0.572±0.003	0.359±0.007	12.55
M1	None	Projection	0.796	0.467	0.651	0.418	0.571±0.003	0.394±0.004	0.00
M2	Global	Projection	0.796	0.493	0.650	0.418	0.574±0.004	0.428±0.004	0.00
M3	Conditional	Projection	0.796	0.501	0.651	0.416	0.575±0.003	0.432±0.004	0.00
M4	None	17-state	0.799	0.468	0.648	0.420	0.571±0.005	0.388±0.003	0.00
M5	Global	17-state	0.796	0.493	0.649	0.423	0.576±0.001	0.430±0.001	0.00
M6	Conditional	17-state	0.784	0.494	0.634	0.418	0.567±0.008	0.427±0.005	0.00

Table 2: seven decision rules, three-seed means ± s.d., document-disjoint

Metric	Contrast	△	95% CI	p_{Holm}
wF1 (official)	M1-M0	-0.001	[-0.006, 0.003]	1.000
	M3-M2	+0.001	[-0.004, 0.006]	1.000
	M4-M1	+0.000	[-0.005, 0.005]	1.000
	M6-M3	-0.008	[-0.018, 0.002]	0.416
	M6-M5	-0.008	[-0.016, -0.001]	0.171
Tuple accuracy	M1-M0	+0.035	[0.028, 0.043]	**0.001**
	M3-M2	+0.004	[-0.001, 0.008]	0.370
	M4-M1	-0.006	[-0.010, -0.002]	**0.028**
	M6-M3	-0.005	[-0.015, 0.004]	0.505
	M6-M5	-0.004	[-0.011, 0.004]	0.505

Table 4: pre-specified paired contrasts, 95% report-cluster bootstrap intervals

- The seven rules span only **0.008** wF1; no official-score contrast survives Holm correction.
- **Adverse result:** 17-state decoding (M4) loses to projection (M1) by **−0.006** tuple accuracy (p_{Holm}=.028).

5 · Replication across four languages

Corpus	Rows	M0 invalid %	N/A net	Subst. net	△ wF1	△ tuple acc.
Chinese (AI CUP)	2,000	12.55	+395	-264	-0.0011	**+0.0350**
English	400	23.17	+148	-86	+0.0032	**+0.0725**
French	400	21.08	+150	-107	+0.0036	**+0.0625**
Japanese	400	31.25	+183	-245	-0.0131	**+0.0708**
Korean	500	22.73	+103	-148	-0.0192	**+0.0293**

Table 7: projection (M1) vs. argmax (M0), one preselected arm per corpus, three-seed means

- Tuple accuracy rises in **32/32** external arms; wF1 in **14/32** (EN 7/8, FR 5/8, JA 0/8, KO 2/8).
- Within-arm directions, not a ranking of languages: the corpora differ in documents, labels, input units, and backbones.

6 · Why the metrics disagree

- Projection's N/A net is positive and its substantive net negative in all five corpora, yet tuple accuracy rises in every one.
- Joint decoding revises PS on **339 of 12,000** rows: 225 become correct, 114 wrong, yet whole-row correctness there falls **114 → 47**.
- Repairing a parent rarely rescues a row whose children stay wrong; breaking a correct parent always loses it.

7 · For audit use

- Validity is a **precondition** for a usable record, not evidence that the record is right.
- A repair can hide an inconsistency, so a deployed validity layer should **log the rows it repaired**.
- Choose the metric by how outputs are used:
  - records → tuple accuracy among valid outputs
  - single-field actions → that field's F1, or recall for a rare class
  - portfolio statistics → class-level scores, keeping validity
  - leaderboards → official wF1, but not for gaps under 0.008

Takeaway

Projection guarantees coherent ESG records without retraining, but it is not a route to a higher official score. For audit use, make validity a precondition and pick the metric that matches the unit at which errors have consequences.

Code and reproduction capsule